

Joel Watlington

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Current Address : Manchester, NH

Online Portfolio : joelwatlington.com

SUMMARY

Highly technical knowledge through **Cloud, DevOps, Web Development, and Automation**

Leadership and communication experience using agile methodology and Jira through user stories, tickets, and Kanban boards

Documentation skills through numerous **Confluence** pages including complete coverage of Elasticsearch, Python, Ansible playbooks, and missile defense software for onsite servers

Currently preparing for the AWS Certified Solutions Architect – Associate while studying professional-level AWS architecture and best practices

QUALIFICATIONS

- Top-secret with SCI Clearance (2021) (exp. Sep. 2028)
- CompTIA Security+ Certification (2025) (exp. Jan. 2028)
- CompTIA Server+ Certification (2025) (Good For Life)
- Given formal congratulations after presenting the first automated test case under NG (2025)
- Nominated for a CARES award at the State of Maine (2023)

EDUCATION

Tennessee Technological University – Cookeville, TN (May 2021)

- Bachelor of Science in Computer Science
- Cumulative GPA: 3.54 Graduated with Cum Laude

Motlow State Community College – Tullahoma, TN (May 2018)

- Associates of Science in Chemistry
- GPA: 3.94

SOFTWARE

- **AWS Cloud** – Architected secure cloud infrastructure utilizing IAM roles, Route 53 DNS routing, and security groups; Engineered automated AMI pipelines, deployed Lambda functions, and managed company wide operations through the AWS console
- **Kubernetes** – Delegated Kubernetes clusters, nodes, and pods to allow for scalability in resources using Docker containers and software while working under Wright Patterson AFB, Northrop Grumman, and the FBI. Engineered scalable microservices, managed pod deployments, developed cybersecurity best practices and configured regular Helm chart upgrades

- **Docker** – Utilized Docker with Docker commands to streamline programming functionality under Wright Patterson AFB, Northrop Grumman, and the FBI to improve Kubernetes resource integration, package control, and a baseline for all deployment operations
- **Jenkins** and **Gitlab CI/CD**, **Agents**, **Gitlab runners**, and **YAML** – Developed pipelines using Gitlab CI/CD and YAML to automate linting, building, testing, and security checking for future deployments and Jfrog storage under Wright Patterson AFB. Crafted large pipeline features under Northrop Grumman to scan every new file for vulnerabilities using Coverity. Additionally supported overnight automation for test cases and post test analysis. Experience readily transferable to **GitHub Actions** due to equivalent workflow and pipeline concepts
- **Ansible** – Guided Ansible playbooks under FBI to dynamically allocate user accounts, streamline patching while also displaying critical version information, and organized health status scripts to replace Puppet licensing through the creation of systemd jobs
- **Terraform** – Designed a completely new terraform structure under FBI so that AWS console implementation would not only be consistent but more secure through the alteration of security groups
- **Linux/Bash** – Actively programmed using bash scripting to automate complex data ingestion pipelines; highly familiar with the Linux operating system to support team development, manage user accounts, and execute system commands including curl, dnf, and scp while working for Northrop Grumman and the FBI
- **SQL Developer**, **PostgreSQL**, **SQL Server Management Studio**, **Amazon RDS**, and **DynamoDB** – Experienced querying relational and NoSQL databases to troubleshoot application issues, retrieve customer data, validate security information, and support development across enterprise cloud environments
- **AWS Lambda** – Integrated serverless automation that executed Ansible patch management workflows on scheduled intervals, eliminating manual maintenance windows and providing automated email notifications upon successful completion.
- **AWS Fargate** – Provisioned containerized build environments to support Infrastructure-as-Code workflows, reducing reliance on persistent EC2 instances while simplifying container deployment and lifecycle management
- **Amazon S3** – Configured S3 buckets to securely store monitoring and infrastructure data in JSON format, enabling downstream integration with enterprise observability platforms
- **Kafka** and **Zookeeper** – Deployed multi-node message brokers and managed Zookeeper states to ensure seamless cluster coordination, topic creation, and fault-tolerant data streaming under FBI
- **Python** – Built multiple scripts i.e. manipulating **JSON** and **NDJSON** files under the Elasticsearch team. Supported python scripting to concisely track the automation flow under Northrop Grumman.
- **Git** and **Bitbucket** – Organized, deployed, and collaborated with multiple different teams across many different repositories through all past work experiences
- **Elastic Stack (Elasticsearch, Kibana, EQL, Lucene, Logstash)** – Built and optimized search queries, dashboards, detection logic, and data ingestion workflows to improve log analysis, threat detection, and operational visibility across enterprise environments

WORK EXPERIENCE

FBI – **Mid-Level Cloud Developer** | **Short contract** | Huntsville, AL (Mar. 2026 – May 2026)

- Engineered an AWS-native AMI build pipeline using AWS **CodePipeline**, **CodeBuild**, **Docker-in-Docker**, and **Packer**, automatically generating hardened Amazon Machine Images for Ansible deployments
- Developed reusable **Terraform** modules to standardize AWS infrastructure, refactoring security groups to replace CIDR-based access with **IAM**-driven role-based permissions that improved security and maintainability
- Automated Linux patch management and system administration by developing **Ansible** playbooks integrated with AWS **Lambda** to enable scheduled patching, health checks, and email notifications without manual intervention. This ultimately generated cost savings by removing Puppet licenses
- Leveraged AWS **Fargate** to provision robust build environments for infrastructure automation, reducing long living EC2 nodes and supporting Infrastructure-as-Code deployment practices
- Designed and deployed a highly available **Kafka** cluster in AWS, configuring Zookeeper coordination, automating topic creation, and managing IAM permissions required for cluster provisioning and lifecycle management
- Built automated monitoring data collection pipelines that gathered infrastructure metrics from multiple APIs, stored JSON artifacts in Amazon **S3**, and supplied data to **Prometheus** and Grafana dashboards for operational visibility
- Worked with Amazon **RDS** and **DynamoDB** to troubleshoot customer support issues, query application data, and validate security-related account information for operational support
- Administered AWS security logging through AWS **Control Tower**, IAM, and CloudTrail to manage user permissions, auditing cloud activity, and supporting secure enterprise cloud operations

Northrop Grumman – **Lead DevOps Automation Engineer** | Huntsville, AL (Sept. 2024 – Feb. 2026)

- Developed and optimized DevOps pipelines through **Jenkins** and **Docker** to ultimately enhance deployment speed and improve project delivery timelines.
- Hands-on experience working with the integration team to produce Ansible playbooks for reliable **Kubernetes** installation and maintenance that facilitated software installation through automated means
- Employed automation software using Eggplant and **Python** code so that test cases and post-test analysis would be generated without any manual intervention
- Orchestrated presentation skills through creating videos to demonstrate competence for government customers. Teaching experience by instructing an entire team how to use automation software
- Assisted Coworkers, Scrum Master, Product Owner, and Technical Lead by creating Jira stories, providing feedback, and aiding in work management as a professional team leader that changed company flow
- Administered user accounts, managed groups, and configured granular sudo privileges with the help of active directory to maintain strict access control environments
- Designed and deployed multiple robust **Bash** scripts to automate complex data ingestion pipelines, reducing manual processing time
- Crafted a large software feature to scan every new file that had been pushed for vulnerabilities using **Coverity**. Additionally supported new pipeline functionality for automation while communicating 24/7 automation to the information systems security manager.

Wright Patterson AFB – **DevOps Engineer** | Dayton, OH

(Mar. 2024 – Aug. 2024)

- Engineered a streamlined **GitLab** pipeline that reduced deployment time by leveraging Docker for secure service container integration
- Optimized **Docker** container images by minimizing package dependencies and debugging any issues using the Linux terminal, successfully reducing image footprint and accelerating cluster pull times
- Utilized docker commands to streamline programming functionality and improve deployment flow, programming control, and a baseline for all operations
- Delegated **Kubernetes** clusters, nodes, namespaces, and pods to allow for scalability in resources for Docker containers and software; engineered scalable microservice architectures, managed pod deployments, and configured cluster ingress controllers
- Streamlined application deployment on Kubernetes clusters by designing and managing dynamic **Helm** charts. Integrated Helm chart deployments into an automated CI/CD pipeline using GitLab
- Consistently navigating **Jfrog** artifactory to solve dependency issues for key libraries and programming Cpp / Hpp files to allow correct mathematical flow

State of Maine – **Software Engineer** | Augusta, ME

(Jun. 2023 – Nov. 2023)

- Enhanced client satisfaction by maintaining **web applications**, resolving bugs, and implementing features to ensure timely communication of expectations using C#, JavaScript, ASP.NET, ASP Classic, and Visual Basic
- Frequently monitoring emails to aid clients regarding software installations, database fixes, or general usage in a timely, polite, and intelligent way
- Succinctly learning cybersecurity and programming techniques throughout over 40 hours of professional training
- Maintained and optimized multiple database platforms, including **MySQL** and **SQL Server**, enhancing application performance and reliability

Robins Air Force Base – **Elasticsearch Engineer** | Warner Robins, GA

(July 2021 – Mar. 2023)

- Assembled custom **Elastic Integration Packages** for clients across the entire ELK stack that improved data accuracy through effective parsing and optimization techniques
- Navigated **Linux** and Windows **PowerShell** administration commands to troubleshoot Elasticsearch problems and aid efficient communication flow
- Architected **Logstash** data pipelines to ingest, prune, and normalize high-volume datasets for accuracy and clean data delivery
- Developed interactive **Kibana dashboards** and deployed Machine Learning to transform raw log data into insightful analysis
- Architected lightweight data collection infrastructure by deploying **Elastic Beats** across all production servers which reduced agent memory consumption
- Engineered, configured, and maintained Elasticsearch production clusters to implement secure node encryption and optimized cluster health configurations to ensure a stable green status
- Enforced automated **ILM policies** across the Elastic Stack to orchestrate data transitions from hot and cold object storage for reducing cloud data retention costs
- Worked a full-stack application using Angular and Spring Boot, enabling efficient data management for government projects across the country